Methodology and verification: Holistic Accuracy

Companion to **Holistic Accuracy and the Limits of Lockean Derivation**, version 7.

Argument and source comparison

The essay tests an unrestricted reading of Dorst's holistic accuracy argument using a score on sets of propositions: separable true/false contributions plus a premium for the truth of their conjunction. Its proofs distinguish the counterexample, marginal-value restrictions, the characterization of maximizing cores, and a separate cross-space invariance theorem.

The published 2019 article's §4.2 and proofs of Propositions 4–5 were compared with those claims. The July 2021 penultimate author draft of Dorst and Mandelkern's Good Guesses was inspected, including its scoring rule, Filtering, belief application and Optionality proof. The comparison does not assume textual identity with the published version.

Verification

The principal record-score example was checked over all 256 three-world belief states using exact rational arithmetic. At credences $(0.70, 0.15, 0.15)$ the unique optimum has score 4.30; at $(0.75, 0.01, 0.24)$ it has score 4.94. The false-belief marginal costs that expose the first gap are 2 and 5 at the two relevant worlds.

A separate exact calculation confirms that the Jamesian rule in Good Guesses, with one fixed parameter $J = 125/64$, produces the same two displayed cores. This establishes overlap in behavior, not identity of the two scoring rules.

The revised existence bridge is Non-vacuity, not permission for every maximizing state. Uniqueness, or agreement of all optima on the relevant proposition, is sufficient for the counterexample. Refinement invariance is now presented as sufficient for a cross-space result. The family $T = 1, F = -n$ on each n-world space verifies that fixed-space Lockeanism alone does not imply it.

Scope and remaining limitations

The essay's positive contribution is an optimizer characterization under strict marginal truth-directedness, not the invention of utility-based guessing or question-sensitive belief. The record score is a counterexample with a proposed motivation, not a complete theory of epistemic value. A stronger interpretation of Dorst's assumptions can exclude it, at the cost of the restrictions the essay identifies.

The proof review and calculations support the stated results; they do not establish exhaustive literature novelty or constitute independent expert review.

Matching files

The manuscript and anonymous PDF share the version 7 text. The source count is **5,035 words**, including title, abstract, headings, mathematical tokens, footnotes and appendix; only the bibliography is excluded. This report concerns argument construction, source fidelity and verification. It is not a reconstruction of a production transcript or a certification of contest eligibility.